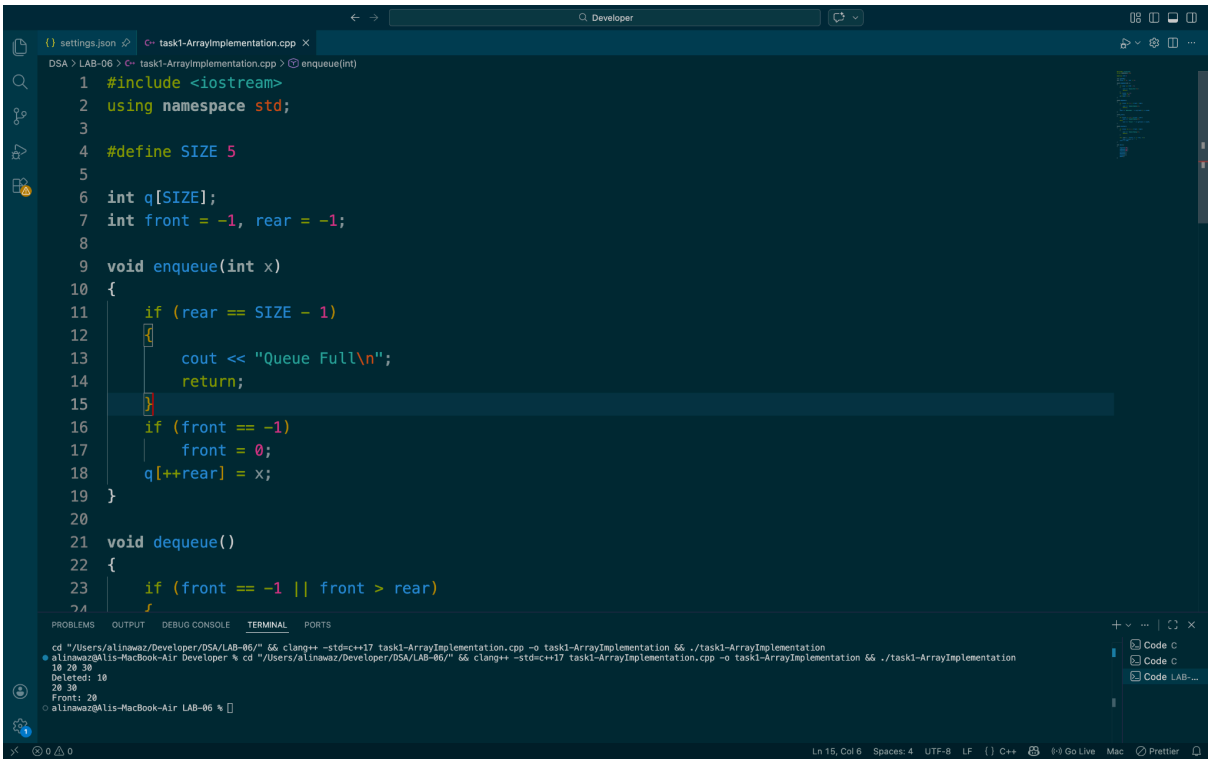
Institute of Geographical Information Systems

CS-250 - Data Structures and Algorithms LAB

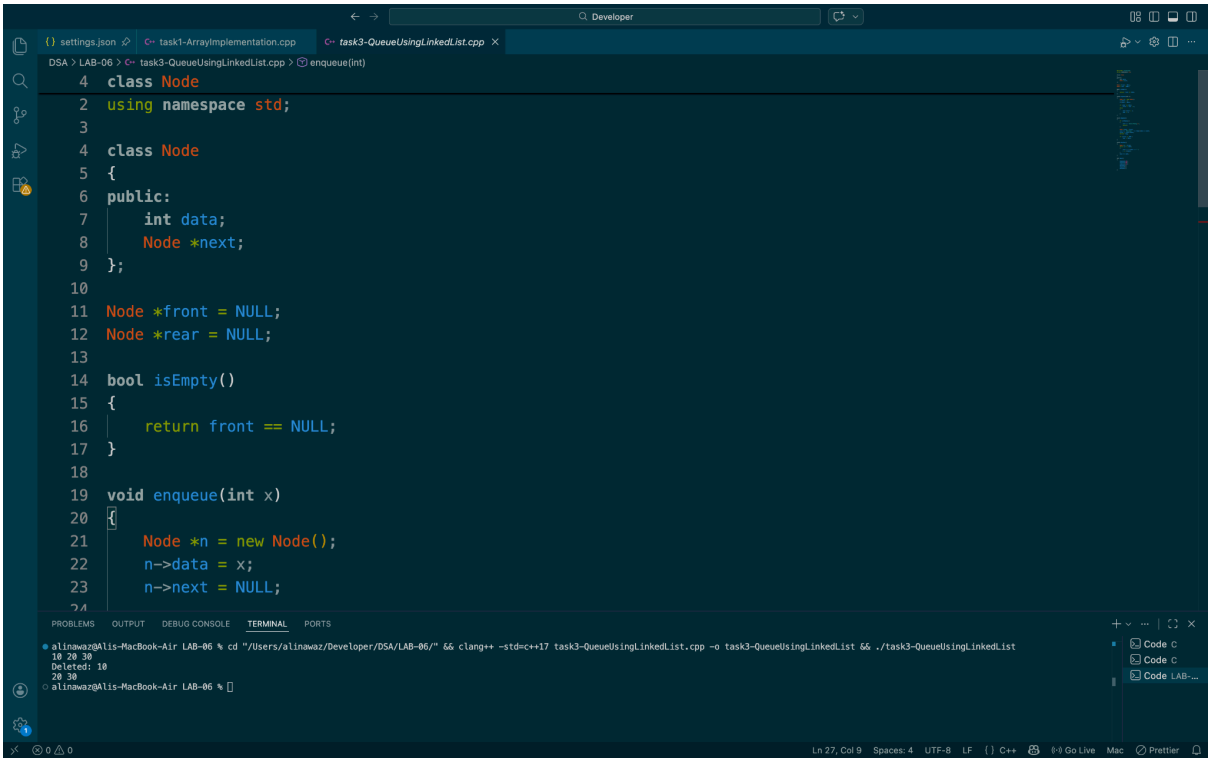
Semester: Spring 2026	**Class: SCEE-IGIS - 2024**
Name: Ali Nawaz	**CMS ID : 00000526123**
Submitted to: Ma'am Alvina Anjum	**Due Date: Feb 25, 2026**

LAB 6: Queue Implementation using Array, Circular Queue & Linked List

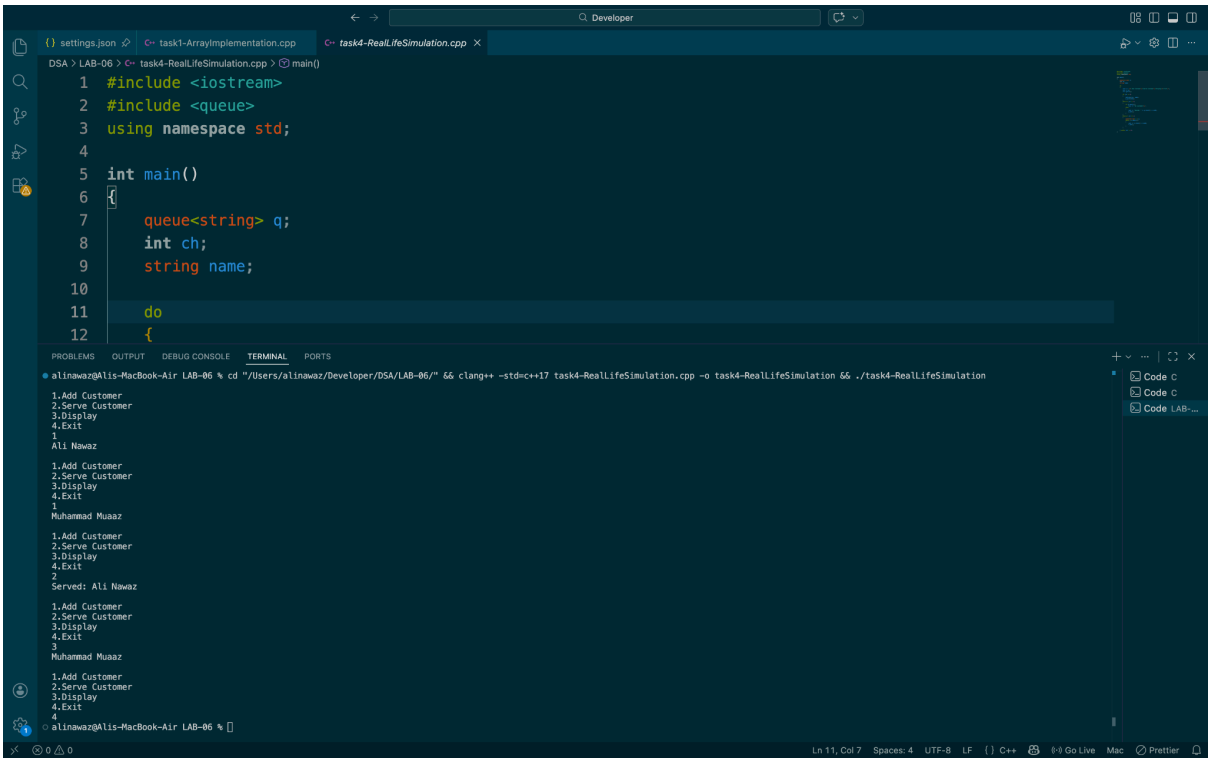
Task 1: Array Implementation (Linear Queue)



Task 2: Queue Using Linked List



Task 3: Real-Life Simulation



Reflection Questions (To Submit)

1. **What is FIFO?**

Ans. FIFO means First In First Out.
The element that enters the queue first is removed first.
Just like people standing in a line.

2. **Why do we need circular queue?**

Ans. In a normal queue, empty spaces are wasted after deleting elements.
Circular queue uses those empty spaces again, so memory is not wasted.

3. **Difference between array and linked list queue?**

Ans. Array Queue:

- Fixed size
- Can become full
- Some memory is wasted

Linked List Queue:

- No fixed size
- Grows when needed
- No memory waste

4. **Where are queues used in operating systems?**

Ans. Queues are used in:

- CPU scheduling
- Process handling
- Printer jobs
- Input/output buffering
- Managing waiting tasks